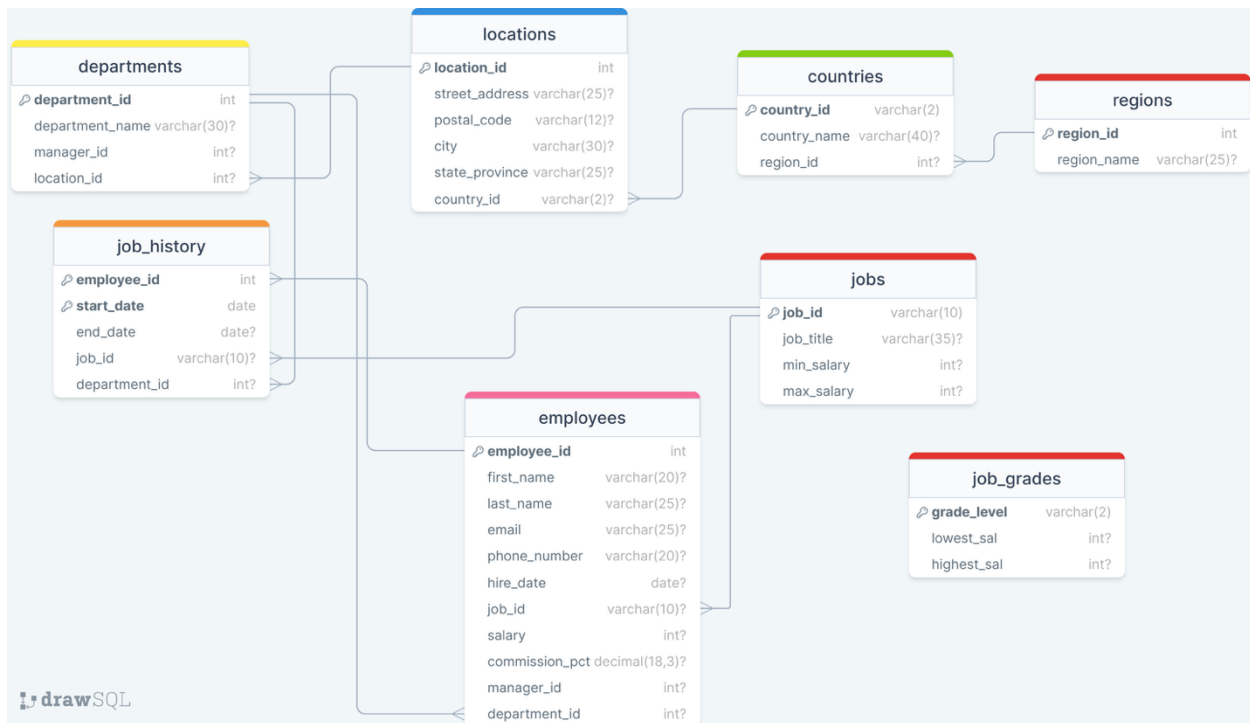


A database for HR is described by the diagram belows.



Tasks:

I. Create tables

1. Write a SQL statement to create a simple table of **countries** including columns **country_id**, **country_name** and **region_id**.
2. Write a SQL statement to create a table named **jobs** including columns **job_id**, **job_title**, **min_salary**, **max_salary** and check whether the **max_salary** amount exceeding the upper limit 25000.
3. Write a SQL statement to create a table named **countries** including columns **country_id**, **country_name** and **region_id** and make sure that no countries except Italy, India and China will be entered in the table.
4. Write a SQL statement to create a table named **countries** including columns **country_id**, **country_name** and **region_id** and make sure that no duplicate data against column **country_id** will be allowed at the time of insertion.
5. Write a SQL statement to create a table named **jobs** including columns **job_id**, **job_title**, **min_salary** and **max_salary**, and make sure that, the default value for **job_title** is blank

and min_salary is 8000 and max_salary is NULL will be entered automatically at the time of insertion if no value assigned for the specified columns.

6. Write a SQL statement to create a table named **countries** including columns country_id, country_name and region_id and make sure that the country_id column will be a key field which will not contain any duplicate data at the time of insertion.

7. Write a SQL statement to create a table of **countries** including columns country_id, country_name and region_id and make sure that the column country_id will be unique and store an auto incremented value.

8. Write a SQL statement to create a table **job_history** including columns employee_id, start_date, end_date, job_id and department_id and make sure that, the employee_id column does not contain any duplicate value at the time of insertion and the foreign key column job_id contain only those values which are exists in the jobs table.

Here is the structure of the table jobs;

Field	Type	Null	Key	Default	Extra
JOB_ID	varchar(10)	NO	PRI		
JOB_TITLE	varchar(35)	NO		NULL	
MIN_SALARY	decimal(6,0)	YES		NULL	
MAX_SALARY	decimal(6,0)	YES		NULL	

9. Write a SQL statement to create a table **employees** including columns employee_id, first_name, last_name, email, phone_number hire_date, job_id, salary, commission, manager_id and department_id and make sure that, the employee_id column does not contain any duplicate value at the time of insertion and the foreign key columns combined by department_id and manager_id columns contain only those unique combination values, which combinations are exists in the departments table.

Assume the structure of departments table below.

Field	Type	Null	Key	Default	Extra
DEPARTMENT_ID	decimal(4,0)	NO	PRI	0	
DEPARTMENT_NAME	varchar(30)	NO		NULL	
MANAGER_ID	decimal(6,0)	NO	PRI	0	
LOCATION_ID	decimal(4,0)	YES		NULL	

10. Write a SQL statement to create a table **employees** including columns employee_id, first_name, last_name, email, phone_number hire_date, job_id, salary, commission, manager_id and department_id and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column department_id, reference by the column department_id of departments table, can contain only those values which are exists in the departments table and another foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table.

Assume that the structure of two tables departments and jobs.

Field	Type	Null	Key	Default	Extra
DEPARTMENT_ID	decimal(4,0)	NO	PRI	0	
DEPARTMENT_NAME	varchar(30)	NO		NULL	
MANAGER_ID	decimal(6,0)	YES		NULL	
LOCATION_ID	decimal(4,0)	YES		NULL	

Field	Type	Null	Key	Default	Extra
JOB_ID	varchar(10)	NO	PRI		
JOB_TITLE	varchar(35)	NO		NULL	
MIN_SALARY	decimal(6,0)	YES		NULL	
MAX_SALARY	decimal(6,0)	YES		NULL	

11. Write a SQL statement to create a table **employees** including columns employee_id, first_name, last_name, job_id, salary and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The specialty of the statement is that, The ON UPDATE CASCADE action allows you to perform cross-table update and ON DELETE RESTRICT action reject the deletion. The default action is ON DELETE RESTRICT.

Field	Type	Null	Key	Default	Extra
JOB_ID	int(11)	NO	PRI	NULL	
JOB_TITLE	varchar(35)	NO			
MIN_SALARY	decimal(6,0)	YES		8000	
MAX_SALARY	decimal(6,0)	YES		NULL	

12. Write a SQL statement to create a table **employees** including columns employee_id, first_name, last_name, job_id, salary and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The specialty of the statement is that, The ON DELETE CASCADE that lets you allow to delete records in the employees(child) table that refer to

a record in the jobs(parent) table when the record in the parent table is deleted and the ON UPDATE RESTRICT actions reject any updates.

Field	Type	Null	Key	Default	Extra
JOB_ID	int(11)	NO	PRI	NULL	
JOB_TITLE	varchar(35)	NO			
MIN_SALARY	decimal(6,0)	YES		8000	
MAX_SALARY	decimal(6,0)	YES		NULL	

13. Write a SQL statement to create a table **employees** including columns employee_id, first_name, last_name, job_id, salary and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The specialty of the statement is that, The ON DELETE SET NULL action will set the foreign key column values in the child table(employees) to NULL when the record in the parent table(jobs) is deleted, with a condition that the foreign key column in the child table must accept NULL values and the ON UPDATE SET NULL action resets the values in the rows in the child table(employees) to NULL values when the rows in the parent table(jobs) are updated.

Field	Type	Null	Key	Default	Extra
JOB_ID	int(11)	NO	PRI	NULL	
JOB_TITLE	varchar(35)	NO			
MIN_SALARY	decimal(6,0)	YES		8000	
MAX_SALARY	decimal(6,0)	YES		NULL	

II. Insert values

1. Write a SQL statement to insert a record with your own value into the table countries against each column.

Here in the following is the structure of the table countries.

Field	Type	Null	Key	Default	Extra
COUNTRY_ID	varchar(2)	YES		NULL	
COUNTRY_NAME	varchar(40)	YES		NULL	
REGION_ID	decimal(10,0)	YES		NULL	

2. Write a SQL statement to insert one row into the table countries against the column country_id and country_name.

Here in the following is the structure of the table countries.

Field	Type	Null	Key	Default	Extra
COUNTRY_ID	varchar(2)	YES		NULL	
COUNTRY_NAME	varchar(40)	YES		NULL	
REGION_ID	decimal(10,0)	YES		NULL	

3. Write a SQL statement to create duplicate of countries table named country_new with all structure and data.

Here in the following is the structure of the table countries.

Field	Type	Null	Key	Default	Extra
-------	------	------	-----	---------	-------

Field	Type	Null	Key	Default	Extra
COUNTRY_ID	varchar(2)	YES		NULL	
COUNTRY_NAME	varchar(40)	YES		NULL	
REGION_ID	decimal(10,0)	YES		NULL	

4. Write a SQL statement to insert NULL values against region_id column for a row of countries table.

5. Write a SQL statement to insert 3 rows by a single insert statement.

6. Write a SQL statement insert rows from country_new table to countries table.

Here are the rows for country_new table. Assume that the countries table is empty.

COUNTRY_ID	COUNTRY_NAME	REGION_ID
C0001	India	1001
C0002	USA	1007
C0003	UK	1003

7. Write a SQL statement to insert one row in jobs table to ensure that no duplicate value will be entered in the job_id column.

10. Write a SQL statement to insert rows into the table countries in which the value of country_id column will be unique and auto incremented.

11. Write a SQL statement to insert records into the table countries to ensure that the country_id column will not contain any duplicate data and this will be automatically incremented and the column country_name will be filled up by 'N/A' if no value assigned for that column.

12. Write a SQL statement to insert rows in the job_history table in which one column job_id is containing those values which are exists in job_id column of jobs table.

13. Write a SQL statement to insert rows into the table employees in which a set of columns department_id and manager_id contains a unique value and that combined values must have exists into the table departments.

14. Write a SQL statement to insert rows into the table employees in which a set of columns department_id and job_id contains the values which must have exists into the table departments and jobs.

III. Update tables

1. Write a SQL statement to change the email column of employees table with 'not available' for all employees.
2. Write a SQL statement to change the email and commission_pct column of employees table with 'not available' and 0.10 for all employees.
3. Write a SQL statement to change the email and commission_pct column of employees table with 'not available' and 0.10 for those employees whose department_id is 110.
4. Write a SQL statement to change the email column of employees table with 'not available' for those employees whose department_id is 80 and gets a commission is less than 20%
5. Write a SQL statement to change the email column of employees table with 'not available' for those employees who belongs to the 'Accounting' department.
6. Write a SQL statement to change salary of employee to 8000 whose ID is 105, if the existing salary is less than 5000.
7. Write a SQL statement to change job ID of employee which ID is 118, to SH_CLERK if the employee belongs to department, which ID is 30 and the existing job ID does not start with SH.
8. Write a SQL statement to increase the salary of employees under the department 40, 90 and 110 according to the company rules that, salary will be increased by 25% for the department 40, 15% for department 90 and 10% for the department 110 and the rest of the departments will remain same.
9. Write a SQL statement to increase the minimum and maximum salary of PU_CLERK by 2000 as well as the salary for those employees by 20% and commission percent by 10.